Complete Curriculum: Quantum Computing Learning App

Course Overview:

Welcome to the Learn section! This curriculum is designed to take you from a curious beginner to someone who understands the physical reality and mathematical foundations of quantum computers. The course is split into three parts:

- **Part 1:** Quantum concepts (the rules of the quantum world).
- **Part 2:** Quantum hardware (how we actually build these machines).
- **Part 3:** Math Companion (optional deep-dives to check your intuition with calculation).

PART 1: Concepts Through Games and Simulations

Module 0: Why Quantum? (Orientation)

The "Aha!" Moment: A quantum computer is a specialist tool, not just a faster laptop.

Lecture Notes:

Quantum computers aren't here to replace your smartphone, make your internet browser faster, or run spreadsheets. Instead, they are highly specialized machines designed to solve specific types of problems that classical computers struggle with.

- **Where they might help:** Simulating quantum systems (like complex molecules for drug discovery or new battery materials), certain optimization and sampling problems, and specific cryptography tasks.
- **Where they won't help:** Everyday computing tasks.
- **The Secret Sauce:** Quantum computers don't just "try every answer at once." They rely on superposition, entanglement, and above all **interference** (arranging amplitudes so wrong answers cancel out and right answers are amplified), followed by a final **measurement**.

Reference Video:

- *Title:* The Map of Quantum Computing - Quantum Computing Explained (Domain of Science)
- *Link:* https://www.youtube.com/watch?v=-UlxHPIEVqA

Quiz:

1. **Which of the following is a task a quantum computer is expected to excel at?**
    - A) Making video streaming faster
    - B) Simulating the behavior of a complex molecule

- C) Running everyday word processing software faster
- D) Storing standard digital photos more efficiently
- **Correct Answer:** B
- **Explanation:** Quantum computers are specialist tools. Because molecules operate on quantum rules, quantum computers are uniquely suited to simulate them. They will not speed up everyday classical tasks like video streaming.

2. **How does a quantum computer find the correct answer to a complex problem?**
   - A) It tries every single possible answer at the exact same time and picks the best one.
   - B) It runs at a much higher clock speed than a regular computer.
   - C) It uses interference to cancel out wrong answers and amplify the probability of the right answer.
   - D) It asks an AI to predict the outcome.
   - **Correct Answer:** C
   - **Explanation:** *Misconception Check!* A quantum computer does NOT try every answer at once. Instead, quantum algorithms use interference—much like noise-canceling headphones—to make the wrong answers cancel each other out, so the right answer is very likely to appear when measured.
3. **Which statement about quantum computers today is most accurate?**
   - A) They have already replaced supercomputers for most scientific work.
   - B) Real machines exist, but they are still noisy; many promising applications need larger, error-corrected machines.
   - C) They can already break all internet encryption.
   - D) They are only a theoretical idea and have never been built.
   - **Correct Answer:** B
   - **Explanation:** *Misconception Check!* You can run circuits on real quantum hardware over the cloud today, but current devices are noisy. Breaking widely used encryption with Shor's algorithm would need far larger, error-corrected machines that do not exist yet.

Module 1: Qubits & Superposition

The "Aha!" Moment: A qubit state is a point on a sphere, not "0 and 1 at the same time."

Lecture Notes:

Classical computers use **bits**, which are strictly 0 or 1. Quantum computers use **qubits**. A qubit's state is described by *amplitudes* over the outcomes 0 and 1.

- **Superposition:** When a qubit is in a superposition, it isn't "0 and 1 at the same time." Rather, it has a probability of being measured as 0, and a probability of being measured as 1.
- **The Bloch Sphere:** Imagine a 3D globe. The North Pole is the state $|0\rangle$, and the South Pole is $|1\rangle$. The equator represents equal-weight superpositions (like $|+\rangle$ and $|-\rangle$), where there is a

50/50 chance of measuring 0 or 1. A qubit's state is just a point on this sphere!

Reference Video:

- *Title:* Quantum Computing in 5 Levels - WIRED (Watch 2:18–6:15)
- *Link:* https://www.youtube.com/watch?v=OWJCfOvochA

Quiz:

1. **If a qubit is in the |+⟩ state (on the equator of the Bloch sphere), what does this mean?**
   - A) It is exactly equal to 0.5.
   - B) It is both 0 and 1 at the exact same time.
   - C) It has equal amplitudes, meaning a 50% chance of being measured as 0 and a 50% chance of being measured as 1.
   - D) The qubit is broken.
   - **Correct Answer:** C
   - **Explanation:** *Misconception Check!* A qubit isn't "both at once." Its state is described by amplitudes, and those amplitudes set the probability of each outcome when it is measured.
2. **On the Bloch sphere, where do the classical states 0 and 1 live?**
   - A) At the North and South poles (|0⟩ and |1⟩).
   - B) On the equator.
   - C) Inside the center of the sphere.
   - D) They constantly orbit the sphere.
   - **Correct Answer:** A
   - **Explanation:** The poles represent the classical, definite states of |0⟩ and |1⟩.
3. **A qubit has a 90% chance of being measured as 0. Where is it on the Bloch sphere?**
   - A) On the equator.
   - B) At the South Pole.
   - C) In the northern hemisphere, close to (but not at) the North Pole.
   - D) Exactly at the North Pole.
   - **Correct Answer:** C
   - **Explanation:** The closer a state is to the North Pole, the more likely the outcome 0. Only a state exactly at the pole gives 0 with certainty; 90% means close to the pole but not on it.
4. **|+⟩ and |−⟩ both give 0 or 1 with 50/50 odds when measured in the standard basis. Are they the same state?**
   - A) Yes, they are identical.
   - B) No. They sit at opposite points on the equator and differ in relative phase.
   - C) No, because |−⟩ has a negative probability.
   - D) Yes, the minus sign is just notation.
   - **Correct Answer:** B
   - **Explanation:** *Misconception Check!* The same probabilities do not mean the same

state. The minus sign is a relative phase: probabilities can never be negative, but amplitudes can. You can tell $|+\rangle$ and $|-\rangle$ apart perfectly by measuring in the X basis (Module 2) or by applying an H gate first (Module 3).

Module 2: Measurement

The "Aha!" Moment: One run is random; thousands of runs reveal the weights.

Lecture Notes:

Measurement is the most aggressive action in quantum mechanics. When you measure a qubit in superposition, its state collapses to a single definite outcome (0 or 1).

- **The Born Rule:** The probability of an outcome is the squared magnitude of its amplitude.
- **Shots & Histograms:** Because one measurement gives a random result, quantum computers run the same circuit thousands of times (called "shots"). The results are plotted on a histogram to reveal the true probability weights.
- **Measurement Bases:** You can look at a qubit from different angles. Measuring in the standard (Z) basis checks if it's North or South. Measuring in the X basis checks the equator. A state that is 100% certain in one basis might be 50/50 random in another!

Reference Video:

- *Title:* 3Blue1Brown - But what is quantum computing? (Grover's Algorithm), first part: state vectors & measurement (0:00–15:52)
- *Link:* https://www.youtube.com/watch?v=RQWpF2Gb-gU

Quiz:

1. **What happens if you prepare a qubit in the exact same superposition state and measure it just once?**
   - A) You will get a fraction, like 0.7.
   - B) You will get either a 0 or a 1, randomly determined by its probability weights.
   - C) You will get 0 and 1 simultaneously.
   - D) It reveals the hidden value it always had.
   - **Correct Answer:** B
   - **Explanation:** You only ever get one definite outcome per measurement. To see the probability distribution, you must run the experiment many times (shots).
2. **True or False: Measurement simply reveals a value that was already hidden inside the qubit.**
   - A) True
   - B) False
   - **Correct Answer:** B
   - **Explanation:** *Misconception Check!* The value is not pre-determined. For example, if you measure the $|0\rangle$ state using an X-basis flashlight, the result is completely 50/50 random. The act of measurement forces the qubit to choose.

3. **A qubit has amplitude 0.6 for $|0\rangle$ and 0.8 for $|1\rangle$. If you run 1,000 shots, roughly how many times will you see 1?**
    - A) About 800
    - B) About 640
    - C) About 360
    - D) About 500
    - **Correct Answer:** B
    - **Explanation:** Born rule: the probability is the amplitude squared, so $P(1) = 0.8^2 = 0.64$, or about 640 out of 1,000. (Check: $0.6^2 + 0.8^2 = 0.36 + 0.64 = 1$.) The exact count changes a little every run; this random spread is called shot noise.
4. **You measure a qubit and get 1. You immediately measure it again in the same basis. What do you get?**
    - A) 1, with certainty
    - B) 0 or 1, with 50/50 odds
    - C) Always 0
    - D) Nothing, because a qubit can only be measured once
    - **Correct Answer:** A
    - **Explanation:** The first measurement collapsed the state to $|1\rangle$. Repeating the same measurement straight away gives the same answer (on an ideal, noise-free qubit).
5. **A qubit in the $|+\rangle$ state is measured in the X basis. What happens?**
    - A) The result is 50/50 random.
    - B) You get the "+" outcome every time.
    - C) You always get 0.
    - D) The qubit is destroyed.
    - **Correct Answer:** B
    - **Explanation:** $|+\rangle$ is one of the X basis's own states, so measuring it in that basis gives a certain result. The same state measured in the Z basis is 50/50. Randomness depends on the basis you choose to measure in.

Module 3: Gates

The "Aha!" Moment: H, Z, H in sequence equals X.

Lecture Notes:

Classical computers use logic gates (AND, OR, NOT). Quantum computers use quantum gates, which are essentially **rotations** of the qubit's point around the Bloch sphere.

- **X Gate:** A bit-flip (rotates North to South, like a classical NOT).
- **Z Gate:** A phase-flip (a half-turn around the vertical Z axis: $|0\rangle$ and $|1\rangle$ stay where they are, while $|+\rangle$ swings around the equator to $|-\rangle$).
- **H Gate (Hadamard):** Moves a state from the poles to the equator (creates a superposition) and vice versa.

- **Reversibility:** Every quantum gate can be undone by applying its inverse.
- **Phase:** Rotating around the equator changes a qubit's *relative phase*. This doesn't change the probability of measuring 0 or 1 right away, but it is the secret ingredient for interference.

Reference Video:

- *Title:* Quantum Country - Quantum computing for the very curious (interactive essay)
- *Link:* https://quantum.country/qcvc
- *Also:* IBM Quantum Learning - Bits, gates, and circuits: https://quantum.cloud.ibm.com/learning/en/courses/utility-scale-quantum-computing/bits-gates-and-circuits

Quiz:

1. **If a qubit is at the North Pole (|0⟩) and you apply an X gate, where does it end up?**
   - A) The equator
   - B) It stays at the North Pole
   - C) The South Pole (|1⟩)
   - D) It collapses
   - **Correct Answer:** C
   - **Explanation:** The X gate is a bit-flip. It rotates the state 180 degrees from |0⟩ to |1⟩.
2. **Why does the Z gate seem to "do nothing" to the probabilities of a |+⟩ state?**
   - A) Because the Z gate is broken.
   - B) Because it only changes the relative phase, moving the state to |−⟩, which still has a 50/50 probability.
   - C) Because Z gates only work on |0⟩.
   - D) Because quantum gates aren't real.
   - **Correct Answer:** B
   - **Explanation:** *Misconception Check!* People think Z does nothing because the probabilities don't change. But it changes the phase (turning |+⟩ into |−⟩). As we will see in the next module, phase is critical for interference!
3. **What do you get if you apply H twice in a row to |0⟩?**
   - A) |1⟩
   - B) |+⟩
   - C) |0⟩
   - D) A random result
   - **Correct Answer:** C
   - **Explanation:** H is its own inverse (H·H = I). The first H moves |0⟩ to |+⟩ on the equator; the second H brings it back to |0⟩. This is reversibility in action.
4. **Starting from |0⟩, you apply H, then Z, then H. What state do you end up in?**
   - A) |0⟩

- B) $|1\rangle$
- C) $|+\rangle$
- D) $|-\rangle$
- **Correct Answer:** B
- **Explanation:** Step by step: H takes $|0\rangle$ to $|+\rangle$, Z takes $|+\rangle$ to $|-\rangle$, and H takes $|-\rangle$ to $|1\rangle$. The whole sequence acts like an X gate (H·Z·H = X), the module's "Aha!" moment. A phase change becomes a bit flip when you sandwich it between two H gates.

Module 4: Phase & Interference

The "Aha!" Moment: A "guaranteed" path can go dark due to cancellation.

Lecture Notes:

This is where quantum computing gets its power. Because quantum states are described by amplitudes (which can be positive, negative, or complex numbers), they can add together like waves.

- **Constructive Interference:** Amplitudes with the same sign add up, making an outcome more likely.
- **Destructive Interference:** Amplitudes with opposite signs cancel each other out, making an outcome less likely, or completely impossible when they cancel exactly.
- **The Mach-Zehnder Interferometer:** If you split a photon's path (putting it in superposition), change its phase, and recombine it, you can direct 100% of the photon to a specific detector. If you block one path, the interference is destroyed, and the photons that get through reach either detector with 50/50 odds.

Reference Video:

- *Title:* Single Photon Interference (Veritasium)
- *Link:* https://www.youtube.com/watch?v=GzbKb59my3U

Quiz:

1. **In a quantum circuit, how do probabilities combine?**
   - A) Probabilities simply add up (e.g., 50% + 50% = 100%).
   - B) They multiply together.
   - C) Amplitudes add together first (allowing for cancellation), and *then* the probability is calculated.
   - D) They average out.
   - **Correct Answer:** C
   - **Explanation:** *Misconception Check!* Probabilities don't just add up. Amplitudes add up first. A positive amplitude and a negative amplitude can cancel out to zero, meaning a seemingly possible outcome becomes impossible (destructive interference).

2. **In a balanced Mach-Zehnder interferometer, one detector never clicks. What happens when you block one of the two arms?**
    - A) The dark detector stays dark.
    - B) The dark detector starts clicking. Photons that get past the block reach either detector with 50/50 odds.
    - C) No photons reach either detector.
    - D) Each photon splits into two half-photons.
    - **Correct Answer:** B
    - **Explanation:** Blocking a path removes one of the two amplitudes, so there is nothing left to cancel against. Blocking a path makes a "forbidden" outcome possible, which is a clear sign that interference, not simple probability, was at work.
3. **Two paths lead to the same detector, each with amplitude of size 1/2. In Case 1 both amplitudes are +1/2. In Case 2 one is +1/2 and the other is −1/2. What is the probability of reaching the detector in each case?**
    - A) 0.5 in both cases (0.25 + 0.25)
    - B) Case 1: 1 (100%); Case 2: 0
    - C) 0.25 in both cases
    - D) Case 1: 1; Case 2: 0.5
    - **Correct Answer:** B
    - **Explanation:** Add the amplitudes first, then square. Case 1: 1/2 + 1/2 = 1, so the probability is $1^2 = 1$. Case 2: 1/2 − 1/2 = 0, so the probability is 0. Adding probabilities instead (0.25 + 0.25) gives the classical answer of 0.5, which experiments contradict.

Module 5A: Entanglement & Bell Tests

The "Aha!" Moment: Correlations that no classical strategy can match.

Lecture Notes:

When two qubits interact (using a two-qubit gate like a **CNOT**), their states can become linked. This is **entanglement**.

- **Exponential Space:** 1 qubit has 2 amplitudes. 2 qubits have 4. 10 qubits have 1024. The state space doubles with every added qubit!
- **Correlations:** Measuring one entangled qubit instantly tells you the state of the other, no matter how far apart they are. But each individual result is still random, so this cannot be used to send a message.
- **Bell's Inequality & CHSH:** Einstein suspected the correlations came from hidden classical instructions ("local hidden variables"). The CHSH game tests this: any classical strategy wins at most 75% of the time, while players sharing entanglement can reach about 85% ($\cos^2(\pi/8) \approx 85.4\%$). Real experiments, recognized by the 2022 Nobel Prize in Physics, beat the classical limit.

Reference Video:

- *Title:* Bell's Theorem: The Quantum Venn Diagram Paradox (MinutePhysics)
- *Link:* https://www.youtube.com/watch?v=zcqZHYo7ONs

Quiz:

1. **Does quantum entanglement allow us to send messages faster than light?**
   - A) Yes, because the other qubit reacts instantly.
   - B) No, because of the no-signaling principle; the outcomes are correlated, but random, so no actual information is transmitted.
   - C) Yes, but only in a vacuum.
   - D) No, because entanglement is an illusion.
   - **Correct Answer:** B
   - **Explanation:** *Misconception Check!* You cannot use entanglement to send an instant text message. The results you get are random. You only see the correlation after comparing notes with the other person using a slow, classical channel.
2. **Two qubits start in $|00\rangle$. You apply H to the first qubit, then a CNOT. What do you see when you measure both qubits many times?**
   - A) Always 00
   - B) 00 or 11, each about 50% of the time; the two qubits always match
   - C) All four outcomes, each about 25% of the time
   - D) Only 01 or 10
   - **Correct Answer:** B
   - **Explanation:** H and CNOT create the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$. Each qubit on its own looks completely random, but the two results always agree. This combination of perfect correlation and individual randomness is the signature of entanglement.
3. **In the CHSH game, what is the best win rate for a classical strategy, compared with a strategy that uses shared entanglement?**
   - A) 50% vs 100%
   - B) 75% vs about 85%
   - C) 85% vs 75%
   - D) 75% vs 100%
   - **Correct Answer:** B
   - **Explanation:** No classical strategy can beat 75%. Entanglement raises the maximum to $\cos^2(\pi/8) \approx 85.4\%$ (the Tsirelson bound). Entanglement is powerful but not magic: it does not let you win 100% of the time.
4. **In general, how many amplitudes do you need to describe the state of 10 qubits?**
   - A) 10
   - B) 20
   - C) 100
   - D) 1,024
   - **Correct Answer:** D
   - **Explanation:** n qubits need 2^n amplitudes, and $2^{10} = 1{,}024$. The count doubles with every

added qubit, which is why simulating around 50 or more qubits on a classical computer becomes extremely hard.

Module 5B: Entanglement as a Resource (Teleportation)

The "Aha!" Moment: You can move a quantum state without copying it.

Lecture Notes:

- **No-Cloning Theorem:** It is physically impossible to make an exact copy of an unknown quantum state. If you try to measure it to copy it, you destroy the superposition.
- **Quantum Teleportation:** How do we move a qubit if we can't copy it? Teleportation! By consuming a shared pair of entangled qubits and sending two classical bits of information, Alice can destroy a quantum state on her end and make it perfectly reconstruct on Bob's end. Bob cannot recover the state until the two classical bits arrive, so teleportation is never faster than light.

Reference Reading:

- *Title:* IBM Quantum Learning - Basics of Quantum Information (see the "Entanglement in action" lesson for teleportation)
- *Link:* https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information

Quiz:

1. **Why can't we just copy a qubit to back up our quantum data?**
   - A) Our hard drives aren't big enough.
   - B) The No-Cloning Theorem states that unknown quantum states cannot be perfectly copied.
   - C) It requires too much electricity.
   - D) Quantum companies patent their qubits so copying is illegal.
   - **Correct Answer:** B
   - **Explanation:** To copy a state, you'd have to measure it to know what to copy. But measuring collapses the superposition, destroying the very state you wanted to copy.
2. **In quantum teleportation, what must Alice send to Bob over a normal (classical) channel?**
   - A) Nothing; the state jumps across instantly.
   - B) Two classical bits: the results of her measurements.
   - C) The original qubit itself.
   - D) A complete list of the state's amplitudes.
   - **Correct Answer:** B
   - **Explanation:** Bob uses the two bits to choose his correction (an X and/or Z gate). Without them, his qubit looks completely random, which is why teleportation cannot send information faster than light. Note that only two bits are sent, not the amplitudes, which could need infinite precision to write down.

3. **After teleportation is complete, what has happened to Alice's original qubit state?**
    - A) She still has an identical copy.
    - B) It was destroyed by her measurement; only Bob now has the state.
    - C) It stays entangled with Bob's qubit forever.
    - D) It physically travelled through space to Bob.
    - **Correct Answer:** B
    - **Explanation:** Teleportation moves a state; it does not copy it. Alice's measurement destroys her copy, so at the end only one copy exists, exactly as the No-Cloning Theorem requires.

Module 6A & 6B: Circuits, Oracles, and Search

The "Aha!" Moment: Algorithms arrange interference so wrong answers cancel.

Lecture Notes:

- **Reading Circuits:** Quantum circuits are read left to right. Lines are qubits, boxes are gates.
- **Oracles:** Think of an oracle as a "black box" puzzle.
- **Deutsch-Jozsa:** Proves that a quantum computer can solve a specific puzzle (is a hidden function constant or balanced?) in exactly one query using interference, while a deterministic classical computer may need up to $2^{n-1} + 1$ queries in the worst case.
- **Grover's Algorithm:** Used for searching. It uses "amplitude amplification"—the oracle flips the sign of the right answer's amplitude, then a diffusion step (a reflection about the average) amplifies it. Repeating this about $(\pi/4)\sqrt{N}$ times makes the right answer the most probable outcome.

Reference Video:

- *Title:* 3Blue1Brown - Grover's Algorithm (15:52–29:30)
- *Link:* https://www.youtube.com/watch?v=RQWpF2Gb-gU
- *Also:* IBM Quantum Learning - Deutsch-Jozsa algorithm: https://quantum.cloud.ibm.com/learning/en/courses/fundamentals-of-quantum-algorithms/quantum-query-algorithms/deutsch-jozsa-algorithm

Quiz:

1. **How does Grover's Algorithm find the right answer in an unsorted database?**
    - A) It scans every item at the exact same time.
    - B) It uses interference to incrementally amplify the amplitude of the correct answer while shrinking the wrong ones.
    - C) It randomly guesses until it gets it right.
    - D) It asks the oracle for the address of the answer.
    - **Correct Answer:** B
    - **Explanation:** Quantum algorithms don't just "try everything." They meticulously

arrange interference over several steps (iterations) to make the correct answer stand out.

2. **In Deutsch-Jozsa, a hidden function is promised to be either constant (the same output for every input) or balanced (0 for exactly half the inputs, 1 for the other half). How many queries does the quantum algorithm need?**
   - A) One
   - B) Two
   - C) Half the inputs plus one
   - D) Every input
   - **Correct Answer:** A
   - **Explanation:** One query plus interference is enough: the amplitudes of the wrong answers cancel. A deterministic classical algorithm may need $2^{n-1} + 1$ queries in the worst case. To be honest about it, a randomized classical algorithm can usually guess correctly with only a few queries, so Deutsch-Jozsa mainly demonstrates interference rather than offering a practical speed-up.
3. **For a search over N = 16 items, about how many Grover iterations give the best chance of success?**
   - A) 1
   - B) 3
   - C) 8
   - D) 16
   - **Correct Answer:** B
   - **Explanation:** The optimal number is about $(\pi/4)\sqrt{N} = (\pi/4)\cdot 4 \approx 3.14$, so 3 iterations. That gives about a 96% chance of finding the marked item. A classical search would need about 8 checks on average.
4. **What happens if you run more Grover iterations than the optimal number?**
   - A) The success probability keeps rising toward 100%.
   - B) The success probability falls again (the algorithm overshoots).
   - C) The algorithm crashes.
   - D) Nothing changes after the peak.
   - **Correct Answer:** B
   - **Explanation:** Each iteration rotates the state by a fixed angle toward the answer. Past the target it rotates away again, so success rises, peaks, and then falls. The Grover explorer shows exactly this rise and overshoot.

Module 7: Noise & Error Correction

The "Aha!" Moment: You can find an error without looking at the data.

Lecture Notes:

Physical qubits are fragile. Heat, radiation, and vibrations cause **decoherence** (losing their

quantum state) and errors (unwanted bit-flips or phase-flips).

- Since we can't copy qubits (No-Cloning), how do we do error correction?
- **Parity Checks:** We use extra qubits (ancilla qubits) to check the *relationship* between data qubits without actually measuring the data itself. Measuring the ancillas gives an error "syndrome" that shows which qubit flipped, without revealing the encoded information.
- **Logical Qubits:** By grouping many noisy "physical" qubits together with error correction codes, we create one far more reliable "logical" qubit. In the 3-qubit repetition code, if each qubit flips with probability p, the logical error rate is $3p^2 - 2p^3$, which is smaller than p whenever p is below 50%.

Reference Reading:

- *Title:* IBM Quantum Learning - Foundations of Quantum Error Correction
- *Link:* https://quantum.cloud.ibm.com/learning/en/courses/foundations-of-quantum-error-correction

Quiz:

1. **In quantum error correction, why don't we just measure the data qubit to see if it flipped?**
    - A) Measuring the data qubit collapses its superposition, destroying the calculation.
    - B) Measuring is too expensive.
    - C) The data qubit is invisible.
    - D) We do measure it, just very quickly.
    - **Correct Answer:** A
    - **Explanation:** *Misconception Check!* Measuring the data directly ruins the quantum state. We have to use clever parity checks to infer errors without looking directly at the data.
2. **In the 3-qubit repetition (bit-flip) code, the parity checks show that qubit 1 disagrees with qubits 2 and 3, which agree with each other. What should you do?**
    - A) Flip qubit 1 back.
    - B) Flip qubits 2 and 3.
    - C) Measure all three qubits directly.
    - D) Throw the result away and start over.
    - **Correct Answer:** A
    - **Explanation:** The syndrome points to the most likely single error: the majority wins. Flipping qubit 1 back restores the encoded state, and you never learn (or disturb) the logical value itself.
3. **Each physical qubit flips with probability p = 0.01. With the 3-qubit repetition code, the logical error rate is $3p^2 - 2p^3$. Roughly what is it?**
    - A) 0.01

- B) 0.03
- C) About 0.0003
- D) Exactly 0
- **Correct Answer:** C
- **Explanation:** $3(0.01)^2 - 2(0.01)^3 = 0.0003 - 0.000002 \approx 0.0003$, about 33 times better than a single qubit. A logical error now needs two or more flips. The code only helps when p is below 50%, and real codes such as the surface code must also handle phase flips.

PART 2: Types of Quantum Computers (Hardware)

Note: All hardware metrics (qubit counts, fidelity) represent the current landscape and are subject to rapid change. Every physical modality represents a different engineering trade-off.

Module 0: What Makes a Good Qubit?

To build a quantum computer, we need a budget.

- **T1 & T2 Times:** T1 is how long a qubit holds its energy before relaxing from $|1\rangle$ to $|0\rangle$; T2 is how long it keeps a stable phase (its superposition). Together they set the coherence time.
- **Gate Speed & Fidelity:** How fast we can do operations, and how accurately.
- **The Budget:** The number of operations you can do is roughly the coherence time divided by the gate time. For example, T2 = 100 µs with 50 ns gates gives a budget of about 2,000 operations, and fewer once gate errors are counted.

Module 1: Superconducting Qubits

- **Who uses them:** IBM, Google, Rigetti.
- **How they work:** Microchips made of superconducting metals (like niobium) cooled to near absolute zero (10-20 millikelvin) in giant dilution refrigerators. They are controlled by microwave pulses.
- **Trade-offs:** Very fast gates, but highly susceptible to noise. The cryogenics are a massive scaling bottleneck.

Module 2: Trapped Ions

- **Who uses them:** IonQ, Quantinuum.
- **How they work:** Individual charged atoms (ions) suspended in a vacuum by electromagnetic fields, controlled by precision lasers.
- **Trade-offs:** Incredible coherence times and all-to-all connectivity within a chain (any ion can interact directly with any other ion in the same trap), but gate speeds are slower than superconducting, and scaling beyond one chain requires shuttling ions or linking separate traps.

Module 3: Neutral Atoms

- **Who uses them:** QuEra, Pasqal, Atom Computing.
- **How they work:** Uncharged atoms gripped by optical tweezers (lasers). Lasers excite them to high-energy Rydberg states; when two atoms are close, an excited atom blocks its neighbor from being excited (the "Rydberg blockade"), which is used to build entangling gates.
- **Trade-offs:** Can pack massive amounts of qubits into tiny arrays, but controlling them accurately at scale is an ongoing physics challenge.

Module 4: Photonic Qubits

- **Who uses them:** PsiQuantum, Xanadu.
- **How they work:** Information is stored in properties of light (photons) traveling through waveguides.
- **Trade-offs:** The waveguides and routing can operate near room temperature (though detectors often need cooling). The main enemy is **loss** (photons escaping or being absorbed) and the probabilistic nature of photon interactions.

Module 5: Silicon Spin Qubits

- **Who uses them:** Intel, Diraq.
- **How they work:** Storing quantum information in the spin of a single electron trapped in a quantum dot on a silicon chip.
- **Trade-offs:** Massive scaling potential because they can be manufactured using existing global semiconductor fab infrastructure. However, tuning millions of microscopic dots perfectly is incredibly difficult.

Module 6: Quantum Annealing

- **Who uses them:** D-Wave.
- **How they work:** Not a "universal" gate-model computer. Instead, you map an optimization problem into a physical energy landscape, let it cool down (anneal), and the system settles into the lowest energy state (the answer).
- **Trade-offs:** Great for specific optimization problems, but cannot run general algorithms like Shor's or Grover's. Whether annealers beat the best classical methods on real problems is still debated.

Module 7: Frontier Ideas

- **Honesty tag:** Experimental/Contested. These approaches are promising but not yet proven at scale.
- **Topological qubits:** Store information in collective properties of exotic materials, so local noise should not easily disturb it. Microsoft is pursuing this approach; early experimental claims are still debated by physicists.
- **Cat (bosonic) qubits:** Encode a qubit in states of a microwave resonator that naturally resist bit-flips, so less error correction is needed (explored by AWS and the startup Alice &

Bob).

- **Modular / networked machines:** Link several smaller quantum processors with photons, instead of building one giant chip.

Module 8 & 9: Physical to Logical / Capstone

No physical qubit is perfect. To run deep algorithms, we group hundreds or thousands of noisy physical qubits into one **Logical Qubit** using surface codes.

- **Trade-off Capstone:** If a client asks you to simulate a molecule deeply, you need high fidelity and logical qubits. If they want to solve a routing puzzle quickly, annealing might work. Architecture is about balancing cooling, speed, noise, and connectivity!

Part 2 Hardware Quiz:

1. **Which hardware modality relies heavily on operating near absolute zero using dilution refrigerators?**
   - A) Photonic
   - B) Trapped Ions
   - C) Superconducting
   - D) Neutral atoms
   - **Correct Answer:** C (Superconducting circuits need millikelvin temperatures to stay superconducting and to keep thermal noise from disturbing the qubits).
2. **What is the main advantage of Silicon Spin qubits?**
   - A) They don't experience decoherence.
   - B) They can be manufactured using the same semiconductor facilities used for standard computer chips.
   - C) They run on light.
   - D) They need no cooling at all.
   - **Correct Answer:** B.
   - **Explanation:** Silicon spin qubits can reuse the chip-making (CMOS) factories behind ordinary processors, which could make scaling to millions of qubits easier. They still need cooling (around 1 kelvin or below) and careful tuning of every quantum dot.
3. **What is the main difference between a physical qubit and a logical qubit?**
   - A) Physical qubits are real; logical qubits are software simulations.
   - B) A logical qubit is made up of many physical qubits working together via error correction to act as one much more reliable qubit.
   - C) A logical qubit is just a faster physical qubit.
   - D) There is no difference; the terms mean the same thing.
   - **Correct Answer:** B.
   - **Explanation:** Error correction spreads one qubit's information across many physical qubits so errors can be detected and fixed. Logical qubits are real, built from real hardware, and they are much more reliable but never perfectly error-free.

4. **A qubit has T2 = 100 µs and each gate takes 50 ns. Ignoring gate errors, roughly how many gates fit in its coherence budget?**
    - A) 20
    - B) 200
    - C) 2,000
    - D) 2,000,000
    - **Correct Answer:** C
    - **Explanation:** 100 µs ÷ 50 ns = 100,000 ns ÷ 50 ns = 2,000 gates. The real budget is smaller because every gate also adds a little error, which is what the qubit metrics lab lets you explore.
5. **What is a key strength of trapped-ion qubits?**
    - A) They have the fastest gates of any modality.
    - B) Long coherence times and all-to-all connectivity within a chain.
    - C) They run at room temperature without lasers.
    - D) They are the easiest to manufacture in standard chip factories.
    - **Correct Answer:** B
    - **Explanation:** Any ion in a chain can interact directly with any other, so fewer SWAP operations are needed. The trade-offs are slower gates than superconducting qubits, and scaling beyond a single chain requires moving ions or linking traps.
6. **In a neutral-atom quantum computer, what lets two atoms interact to perform an entangling gate?**
    - A) Their electric charge
    - B) The Rydberg blockade
    - C) Superconducting wires between them
    - D) A fiber-optic cable
    - **Correct Answer:** B
    - **Explanation:** The atoms are neutral (charge is what trapped ions use). Lasers excite an atom to a high-energy Rydberg state, and a nearby atom then cannot be excited. This conditional behavior is the basis of the entangling gate.
7. **What is the main enemy of photonic quantum computers?**
    - A) Heat inside the waveguides
    - B) Photon loss
    - C) Stray magnetic fields
    - D) Slow gate speeds
    - **Correct Answer:** B
    - **Explanation:** Every lost photon is lost information, and the chance that everything works falls exponentially with the number of lossy components (see the loss-budget lab). Honesty note: the waveguides can run near room temperature, but the single-photon detectors usually need cryogenic cooling.
8. **Which statement about quantum annealers is correct?**
    - A) They are universal gate-model quantum computers.

- B) They are specialized for optimization problems and cannot run general gate-model algorithms like Shor's.
- C) They have proven exponential speed-ups for all optimization problems.
- D) They do not use any quantum effects.
- **Correct Answer:** B
- **Explanation:** An annealer maps a problem onto an energy landscape and lets the system settle toward the lowest-energy state. It is a different kind of machine from gate-model computers, and whether it beats the best classical methods is still an open debate.

9. **Which honesty tag best fits topological qubits today?**
   - A) Accurate
   - B) Analogy
   - C) Experimental/Contested
   - D) Simplified model
   - **Correct Answer:** C
   - **Explanation:** Topological qubits promise built-in protection from noise, but the experimental evidence is still debated, so they belong under Experimental/Contested.

PART 3: Math Companion (Optional)

For those who want to see the numbers behind the simulations.

Week 0: Primer

- **Vectors:** A list of numbers. Quantum states are column vectors.
- **Matrices:** Grids of numbers used to multiply and transform vectors.
- **Complex Numbers:** Numbers with a real and imaginary part ($i = \sqrt{-1}$), allowing for magnitude and phase.

Lesson 1: State Vectors & Dirac Notation

- The state $|0\rangle$ is written as the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
- The state $|1\rangle$ is written as the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
- A superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is written as $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$.
- **Rule:** To conserve 100% probability, $|\alpha|^2 + |\beta|^2 = 1$. (Normalization).

Lesson 2: Gates as Matrices

Quantum gates are Unitary Matrices. When a gate acts on a qubit, you multiply the matrix by the state vector.

- **X Gate Matrix:** $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- **H Gate Matrix:** $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
- *Challenge Check:* If you multiply the X matrix by the $|0\rangle$ vector, you get the $|1\rangle$ vector!

Lesson 3: Multiple Qubits

To combine qubits, we use the **Tensor Product**.

- Two qubits mean a vector of 4 amplitudes: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- A 2-qubit gate like CNOT is a 4x4 matrix.

Lesson 4: Measurement, Phase, and Interference Math

- The Mach-Zehnder interferometer is represented mathematically as $H \cdot Phase \cdot H$.
- If you multiply these matrices, you will see mathematically how the negative components of the matrix cancel out specific amplitudes (destructive interference), proving why one detector goes dark.

Lesson 5: Grover's Algorithm in 2D

- Grover's algorithm is just geometry! The Oracle is a reflection matrix (flipping the sign of the target state). The Diffusion operator is a reflection around the average.
- Optimal iterations: Roughly $\frac{\pi}{4}\sqrt{N}$, where N is the number of items.

Part 3 Math Quiz:

1. **If a quantum state vector is $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, what must be true for it to be a valid, normalized state?**
   - A) $\alpha + \beta = 1$
   - B) $|\alpha|^2 + |\beta|^2 = 1$
   - C) $\alpha = \beta$
   - **Correct Answer:** B (The sum of the squared magnitudes of the amplitudes must equal 1, representing 100% total probability).

2. **If you apply the X gate matrix to the vector $\begin{pmatrix}1\\0\end{pmatrix}$, what is the result?**
   - A) $\begin{pmatrix}0\\1\end{pmatrix}$
   - B) $\begin{pmatrix}1\\1\end{pmatrix}$
   - **Correct Answer:** A (The matrix multiplication flips the top and bottom values, turning $|0\rangle$ into $|1\rangle$).
3. **A qubit is in the state (1/√2)|0⟩ + (i/√2)|1⟩. What is the probability of measuring 1? (Lesson 1)**
   - A) i/√2
   - B) −1/2
   - C) 1/2
   - D) 0
   - **Correct Answer:** C
   - **Explanation:** Probability is the squared magnitude: $|i/\sqrt{2}|^2 = |i|^2/2 = 1/2$. A complex phase like i changes the state but not this probability, and a probability can never be complex or negative.
4. **Using the H gate matrix (1/√2)[[1, 1], [1, −1]], what is H applied to |0⟩ = (1, 0)? (Lesson 2)**
   - A) (1/√2)(1, 1), which is |+⟩
   - B) (1/√2)(1, −1), which is |−⟩
   - C) (0, 1), which is |1⟩
   - D) (1, 0), which is |0⟩
   - **Correct Answer:** A
   - **Explanation:** Multiplying a matrix by (1, 0) picks out its first column: (1/√2)(1, 1) = |+⟩. Try H on (0, 1) yourself: you should get |−⟩.
5. **Which property must every quantum gate matrix U have? (Lesson 2)**
   - A) All of its entries are real numbers.
   - B) It is unitary: $U^\dagger U = I$.
   - C) Its determinant is zero.
   - D) It is symmetric.
   - **Correct Answer:** B
   - **Explanation:** Unitarity keeps every state normalized (total probability stays 1) and guarantees the gate can be undone by $U^\dagger$. Entries may be complex (for example, the S gate contains i), and a zero determinant would make the gate irreversible.
6. **Using the order |00⟩, |01⟩, |10⟩, |11⟩, what is the 4-amplitude vector for the two-qubit state |01⟩? (Lesson 3)**
   - A) (1, 0, 0, 0)

- B) (0, 1, 0, 0)
- C) (0, 0, 1, 0)
- D) (0, 0, 0, 1)
- **Correct Answer:** B
- **Explanation:** Tensor product: $|0\rangle \otimes |1\rangle = (1, 0) \otimes (0, 1) = (1\cdot0, 1\cdot1, 0\cdot0, 0\cdot1) = (0, 1, 0, 0)$. The single 1 sits in the $|01\rangle$ slot.

7. **In the Mach-Zehnder model H · Phase · H with phase $\phi = 0$, a photon enters in $|0\rangle$. Where does it exit? (Lesson 4)**
   - A) Port $|0\rangle$, with certainty
   - B) Port $|1\rangle$, with certainty
   - C) Either port, 50/50
   - D) It is absorbed
   - **Correct Answer:** A
   - **Explanation:** With $\phi = 0$ the phase gate is the identity, so $H \cdot H = I$ and the photon always leaves through port $|0\rangle$; the other detector stays completely dark. With $\phi = \pi$ the phase gate becomes Z, and $H \cdot Z \cdot H = X$ sends every photon to port $|1\rangle$ instead.
8. **About how many Grover iterations are needed to search N = 100 items? (Lesson 5)**
   - A) 3
   - B) 8
   - C) 50
   - D) 100
   - **Correct Answer:** B
   - **Explanation:** $(\pi/4)\sqrt{100} = (\pi/4)\cdot10 \approx 7.85$, so about 8 iterations. A classical search needs about 50 checks on average, which shows Grover's quadratic speed-up.